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|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | the X-rays <u>must</u> change if the maximum speed of the electrons decreases, the minimum wavelength of the X-rays increases, and vice versa.                                |
| 29 | B | Option B is the de facto statement of the Heisenberg uncertainty principle. Option A is merely the mathematical relationship of the principle. Options C and D are incorrect. |
| 30 | A | From the decay law, $\frac{dN}{dt} = -\lambda N$ , where $\frac{dN}{dt}$ is the rate of decay of nuclei in the                                                                |